$$\begin{bmatrix} 39 & 39 & 126 & 126 \\ 39 & 39 & 126 & 126 \\ 39 & 39 & 126 & 126 \\ 39 & 39 & 126 & 126 \end{bmatrix}$$

CSVTUonline.com

Dictionary contains 256 characters from 0-255.

(*d*) What are different lossy compression techniques? Explain any two in detail. 7

5. (*a*) What are chain code and shape number? 2

(*b*) What are objectives of image representation techniques? What kind of information we can use to represent any image? Explain in detail. 7

(*c*) Explain polygon approximation techniques of image description with suitable example. 7

(*d*) Explain need of co-occurrance matrix. Write down steps to generate a co-occurrence matrix of image, given in figure-3, where position operator Q defined as 'one pixel immediately to the right'. 7

1	1	7	5	3	2
5	1	6	1	2	5
8	8	6	8	1	2
4	3	4	5	5	1
8	7	8	7	6	2
7	8	6	2	6	2

Figure-3

CSVTUonline.com

333672(33)

BE (6th Semester)
Examination, April - May, 2016

(New Scheme)

CSVTUonline.com

Image Processing

Time Allowed : 3 hours — *Maximum Marks* : 80

Minimum Pass Marks : 28

CSVTUonline.com

Note : (*i*) Part (*a*) of each question is compulsory. Attempt any two parts from (*b*), (*c*) and (*d*).

(*ii*) The figures in the right-hand margin indicate marks.

1. (*a*) What is false contouring? 2

(*b*) Explain image formation model with diagram. Differentiate between linear versus non-linear operation on image. 7

(*c*) What is difference between saturation and noise? How is image averaging useful to enhance image quality? 7

(*d*) Suppose that a 3-bit image ($L=8$) of size 64×64 pixel has the intensity distribution shown in table-1. Write a procedure for histogram equalization of the given intensity distribution : 7

CSVTUonline.com

r_k	n_k
$r_0=0$	790
$r_1=1$	1023
$r_2=2$	850
$r_3=3$	656
$r_4=4$	329
$r_5=5$	245
$r_6=6$	122
$r_7=7$	81

Table - 1

CSVTUonline.com

2. (*a*) How can we convert spatial domain to frequency domain? 2

(*b*) Why is Marr-Hildreth edge detector preferred over any other edge detector? Explain in detail. 7

(*c*) Explain global processing, using hough transform with suitable example. 7

(*d*) Explain first order and second order derivatives and its properties in detail. 7

3. (*a*) What kind of logical operation we can apply in any binary image? 2

(*b*) Explain algorithm of boundary extraction and region filling with example. 7

(*c*) What is meant by, dialation and erosion are increasing transformations? How is connected component extracted from any image using morphology operations? 7

(*d*) What are opening and closing operations of morphology? Explain opening and closing algorithm for following image and structuring element given in figure-1 and figure-2. 7

CSVTUonline.com

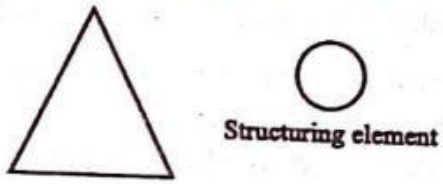

Figure-1 : Apply opening technique

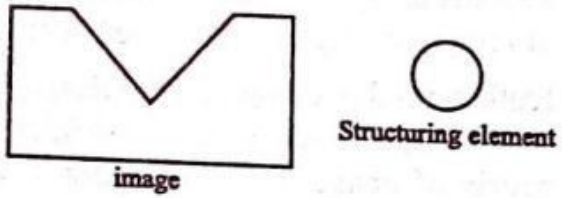

Figure-2 : For closing operation

4. (*a*) What do you mean by redundancies in image? 2

(*b*) Explain the need of image compression. How is run length encoding used for compression? Is it lossy? Justify your answer. 7

CSVTUonline.com

(*c*) Write down LZW compression code for given 4×4 pixel, 8-bit image. 7